Akoizo Original Practice - Form 1 - Answer key

Division B | Circuit Lab | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) A negative charge's force is…

Unit: Electrostatics, historical ideas, and event reasoning.
Worked-case data: Two point charges keep their values while separation triples. Predict force.

A. Opposite the electric field

B. Always with the field

C. Always zero

D. Independent of its charge

Correct: A

2. (2 points) An LED's forward voltage is…

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

A. Device- and condition-dependent

B. Always 0.7 V

C. Always zero

D. Equal to any source

Correct: A

3. (2 points) A voltmeter connects…

Unit: Breadboards, multimeters, and a first virtual build.
Worked-case data: A known 100 Ω resistor has 2 V across it; an unknown in series has 4 V. Find the unknown.

A. Across two nodes

B. As an intentional source short

C. Only instead of every wire

D. Only to ground alone

Correct: A

4. (2 points) Tripling distance changes force to…

Unit: Electrostatics, historical ideas, and event reasoning.
Worked-case data: Two point charges keep their values while separation triples. Predict force.

A. One ninth

B. Three times

C. One third

D. Nine times

Correct: A

5. (2 points) A series resistor mainly…

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

A. Limits current

B. Creates charge

C. Guarantees any voltage is safe

D. Reverses polarity

Correct: A

6. (2 points) An ohmmeter generally requires…

Unit: Breadboards, multimeters, and a first virtual build.
Worked-case data: A known 100 Ω resistor has 2 V across it; an unknown in series has 4 V. Find the unknown.

A. A de-energized suitable circuit

B. A live mains circuit

C. Maximum possible voltage

D. No instrument instructions

Correct: A

7. (2 points) Neutral-object attraction can arise from…

Unit: Electrostatics, historical ideas, and event reasoning.
Worked-case data: Two point charges keep their values while separation triples. Predict force.

A. Polarization

B. Creation of protons

C. Loss of charge conservation

D. Only gravity

Correct: A

8. (2 points) A diode is generally…

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

A. Nonlinear

B. An ideal fixed resistor

C. Always conducting both ways

D. A mechanical switch

Correct: A

9. (2 points) Divider inference needs…

Unit: Breadboards, multimeters, and a first virtual build.
Worked-case data: A known 100 Ω resistor has 2 V across it; an unknown in series has 4 V. Find the unknown.

A. The same series current

B. Equal resistor values

C. Zero source voltage

D. No known component

Correct: A

10. (4 points) Distinguish field and potential.

Unit: Electrostatics, historical ideas, and event reasoning.
Worked-case data: Two point charges keep their values while separation triples. Predict force.

answer: Field is a vector force-per-charge; potential is scalar energy-per-charge. (4 points).

11. (4 points) Why declare the diode model?

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

answer: Ideal, constant-drop, and detailed models predict different currents. (4 points).

12. (4 points) Why check rail continuity?

Unit: Breadboards, multimeters, and a first virtual build.
Worked-case data: A known 100 Ω resistor has 2 V across it; an unknown in series has 4 V. Find the unknown.

answer: Breadboard rails may be split or connected differently from the assumed layout. (4 points).

13. (4 points) Why convert units first?

Unit: Electrostatics, historical ideas, and event reasoning.
Worked-case data: Two point charges keep their values while separation triples. Predict force.

answer: The constant assumes compatible units; prefixes and squared distance otherwise produce major errors. (4 points).

14. (4 points) Name two LED faults.

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

answer: Reversed orientation and an open path; node measurements can distinguish them. (4 points).

15. (4 points) What does meter loading mean?

Unit: Breadboards, multimeters, and a first virtual build.
Worked-case data: A known 100 Ω resistor has 2 V across it; an unknown in series has 4 V. Find the unknown.

answer: The instrument’s finite impedance changes the quantity being measured. (4 points).

16. (4 points) Why is Ah not sufficient for battery energy?

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

answer: Voltage and operating conditions are also required. (4 points).

17. (4 points) Why is logical one not always 5 V?

answer: Physical signal ranges depend on device technology and supply. (4 points).

18. (4 points) How do you validate an SPDT arrangement?

Unit: LED matching, charge-time estimates, and troubleshooting.
Worked-case data: Estimate ideal time for 1.5 Ah supplied at constant 0.30 A.

answer: Trace continuity for every switch combination with labeled common and traveler contacts. (4 points).